

A 27-input sorting network with 152 comparators and 13 layers, and exact negative results for depth 13 on 30 and 32 inputs

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Abstract

We present a comparator network on 27 channels with 152 comparators and 13 layers. It improves the size of the best known 13-layer 27-input network on Dobbelaere’s list of sorting networks from 153 to 152 at equal depth. The network is the published 153-comparator network with a single comparator, (23, 26) in layer 7, deleted. That comparator is not redundant in the classical sense—it changes some binary inputs—so standard redundancy stripping does not find it; an exact removability test does. Correctness was established by two independently written exhaustive verifiers over all $2^{27} = 134,217,728$ binary inputs and confirmed by a third algorithm (output-set propagation). An exhaustive single-comparator deletion test over all 54 networks of the list finds no other removable comparator, and no pair of comparators can be removed from the new network or from the 28-input 159-comparator 13-layer network.

The network was found while attacking the problem that motivated the work: a 13-layer sorting network on 29, 30, 31 or 32 channels, for which the best known depth is 14. That attack failed, and we document how. Wang’s 2025 construction of a 13-layer 28-input network (stacked reflection-symmetric block prefixes, a greedy sixth layer, SAT for layers 7 to 13) was reproduced end to end and transferred faithfully to 30 channels (16+14 blocks) and 32 channels (16+16 blocks and the 5-cube). For every one of the 28 six-layer prefixes tested—22 on 30 channels from three families and 6 on 32 channels from two families—the SAT instance for a reflection-symmetric completion in seven further layers is unsatisfiable, each proof taking between 500 and 1800 seconds with MiniSat; one instance was re-proved without the optional normal-form constraints. We also show that Wang’s 159 comparators are optimal for his first prefix among reflection-symmetric completions, and that a greedy seventh layer destroys completability even for Wang’s own completable prefixes, so fast negatives of that kind carry no information. Depth 13 for 29 to 32 inputs remains open; we give a ranked list of what a follow-up should try. All code, networks, run logs and verification transcripts are in the accompanying repository.

1 Introduction

A comparator network on n channels is a fixed sequence of compare–exchange operations; it is a sorting network if it sorts every input. Two cost measures are classical: the size (number of comparators) and the depth (number of parallel layers, each layer a set of comparators on pairwise disjoint channels). Optimal values are known only for small n : optimal depth up to $n = 17$ [5, 4] and optimal size up to $n = 12$ [7]. Beyond that, the state of the art is a table of best known upper bounds maintained by Dobbelaere [2], whose entries come from classical constructions [9, 1], from heuristic search (SorterHunter [10]) and from SAT-based prefix completion [6, 3].

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At the time of this work (list dated 7 November 2025) the depth entries for 25 to 32 inputs read: 25, 26, 27 and 28 inputs at depth 13, and 29, 30, 31 and 32 inputs at depth 14, with lower bounds of 10 for all of them. The depth-13 entries for 27 and 28 are credited to Wang [3], who found a reflection-symmetric 28-input network with 159 comparators in 13 layers; the depth-14 entries for 29 to 32 come from Batchier’s odd–even merge [9]. The question we set out to answer was whether Wang’s method extends to 29 to 32 inputs, which would lower four table entries by one layer at once: a 32-input network of depth 13 yields 31-, 30- and 29-input networks of the same depth by channel deletion.

Contributions.

1. A 27-input sorting network with 152 comparators and 13 layers (Section 6), verified by two independent exhaustive checks, improving the list entry (153, 13).
2. An exhaustive single-comparator deletion analysis of all 54 networks on the list (Section 6.4): exactly one comparator anywhere on the list is deletable, namely the one above. Pair deletions from the 27/152 and 28/159 networks are impossible.
3. A size-optimality statement for Wang’s prefix (Section 6.5): no reflection-symmetric 13-layer completion of his first six-layer prefix has fewer than 76 suffix comparators, so 159 is optimal for that prefix under symmetry.
4. Exact negative results for the transfer of Wang’s construction to 30 and 32 channels (Sections 7 and 8): 28 six-layer prefixes from five families, all with unsatisfiable seven-layer completion instances, plus a control run showing that the verdict does not depend on the normal-form constraints.
5. A methodological observation (Section 5.3): adding a greedy seventh layer to Wang’s own eight completable 28-channel prefixes makes all eight instances unsatisfiable in under ten seconds. Greedy minimum-output extension beyond layer 6 is therefore the wrong tool, and negatives obtained that way say nothing about the underlying six-layer prefixes.
6. A complete, dependency-light reimplementation of the pipeline (prefix generation with symmetric subsumption, stacking, greedy extension, the SAT encoding of [4] with additional controls, decoding with untangling, stripping, re-layering), two independent verifiers with a test suite, and every run’s parameters and log, so that each claim can be rerun.

Rules of evidence. Everything labelled VERIFIED passed both verifiers on all 2^n binary inputs. Everything labelled NEGATIVE is an exact statement about specific prefixes under a stated completion model, backed by a solver’s proof of unsatisfiability. Everything that timed out, was killed or was not run is labelled INCONCLUSIVE. A timeout is never treated as evidence of nonexistence, and no running time or table value in this paper was adjusted from what the logs record.

2 Preliminaries

Networks. Channels are numbered 0 to $n - 1$. A comparator (i, j) with $i < j$ replaces the values on channels i and j by their minimum on i and their maximum on j . A layer is a set of comparators on pairwise disjoint channels; a network of depth d is a sequence of d layers. Its size is the total number of comparators. Following Dobbelaere’s list, a network is written one layer per line as a list of pairs.

Zero–one principle. A network sorts all inputs from a totally ordered set if and only if it sorts all 2^n binary inputs [1, §5.3.4]. All verification in this paper is over binary inputs.

Output sets. For a prefix P (the first k layers) let $\text{out}(P)$ be the set of binary vectors P produces from all 2^n binary inputs. A suffix S completes P to a sorting network if and only if

S sorts every vector in $\text{out}(P)$. The cardinality $|\text{out}(P)|$ is the standard measure of a prefix’s quality: smaller sets give smaller SAT instances and, empirically, are easier to complete. The output set of a first layer with m comparators has $3^m 2^{n-2m}$ elements. Two reference values are Dedekind numbers: a 4-cube on 16 channels (four layers of dimension-wise comparators) has 168 outputs and a 5-cube on 32 channels has 7581.

Reflection symmetry. The reflection $\rho(i) = n - 1 - i$ maps a comparator (i, j) to $(\rho(j), \rho(i))$. A network is reflection-symmetric if every layer is mapped to itself. Symmetry halves the number of free variables in a SAT encoding; it is the restriction Wang used, and the one under which every negative result in this paper is stated. A network on an odd number of channels cannot be reflection-symmetric layer by layer, since a comparator on the middle channel has no disjoint mirror image; this is why a direct search on 29 channels needs a relaxed symmetry and why 29- and 31-input networks are usually obtained from 30- and 32-input ones by channel deletion.

Channel deletion. Fixing the input of the last channel to $+\infty$ (or of the first to $-\infty$) makes every comparator on that channel a pass-through, so removing those comparators and renumbering gives a sorting network on $n - 1$ channels of the same depth and smaller size [1, §5.3.4]; an interior channel can be deleted in the same way after relabelling channels along the path the fixed value takes.

Redundant and removable comparators. A comparator is *redundant* if it never swaps its inputs on any output of the preceding prefix; redundant comparators are removed by standard stripping. A comparator can, however, swap inputs on some prefix outputs and still be unnecessary for sorting, because later comparators repair the difference. We call such a comparator *removable*. Removability must be decided by an exact test (does the network without it still sort?), which is what Section 6 does. Completeness of a prefix is not monotone under adding comparators [1, §5.3.4, exercise 21]: adding comparators can destroy the possibility of completing a prefix within a given depth, which is the mechanism behind Section 5.3.

Re-layering. A network’s depth is the number of layers after greedy re-layering (each comparator placed in the earliest layer after the last comparator sharing one of its channels). All depths reported here were obtained this way, and all sizes were reported after stripping.

3 Status of the problem

Before any computation, the state of the problem was established from primary sources on 20 September 2026.

- Dobbelaere’s list was last updated on 7 November 2025 (page footer). The served page is byte-identical to the head of the site’s repository, whose only later commit (22 February 2026) changed median-selection networks. Rows: 29 inputs (164, 15), (166, 14); 30 inputs (172, 14); 31 inputs (180, 14); 32 inputs (185, 14); depth bounds 10 to 14 for all four. Rows for 26, 27 and 28 inputs: (138, 15), (139, 14), (141, 13); (147, 16), (148, 14), (153, 13); (155, 14), (159, 13). The depth-13 entries for 27 and 28 are credited to Wang [3], the one for 26 to SorterHunter.
- SorterHunter’s published network directory [10] contains, for 29 to 32 inputs, only depth-14 and depth-15 networks.
- Searches of arXiv (API and full text), Semantic Scholar, OpenAlex, Google Scholar and GitHub found no claim of depth 13 for 29 to 32 inputs. Wang’s paper [3] (v1 6 November 2025, v2 22 November 2025, no later version) had no citations, and its own Table 1 lists 29 to 32 inputs at upper bound 14. No lower-bound argument in the literature comes within three layers of excluding depth 13 for these sizes.

Depth 13 was therefore open for 29, 30, 31 and 32 inputs when the work started.

4 Methods

4.1 Two independent verifiers

Verifier A (`src/verify_c.c`, C with OpenMP) is a bit-sliced simulator. Sixty-four binary inputs are packed into one 64-bit word per channel (the low six bits of the input index select the lane; the remaining $n - 6$ bits are constant across a word), every comparator becomes one AND and one OR, and an output word is sorted if and only if for every adjacent pair of channels $w_c \wedge \neg w_{c+1}$ is zero. All 2^{n-6} words are processed, and the number of failing inputs and the first failing input are reported. It exits with status 0 if the network sorts, 1 if not, 2 on a parse error.

Verifier B (`src/verify_py.py`, Python 3 and numpy) has its own parser and two algorithms unrelated to Verifier A’s. Mode `exhaustive` simulates all 2^n inputs as integers in chunks of 2^{22} and tests each output for the form $0^a 1^b$. Mode `outputset` computes the set of reachable binary vectors layer by layer (product construction after the first layer, then comparator application and deduplication) and checks that the final set consists exactly of the $n + 1$ sorted vectors; it also reports the size of the reachable set after every layer. The option `-mode all` runs both.

Tests (`tests/test_verifiers.py`, four tests). Both verifiers accept every network of Dobbe-laere’s list with $n \leq 20$ and reject each of those networks with its last comparator deleted; both accept Wang’s 28-input network and reject it with one comparator removed; both reject malformed inputs with a parse error, accept the one-comparator 2-sorter and a three-layer 4-sorter, and reject the same 4-channel network without its last layer. A second suite (`tests/test_pipeline.py`, two tests) exercises the search pipeline of Section 4.2: it reproduces Wang’s counts of non-redundant reflection-symmetric 12-channel prefixes (41 at depth 2, 1502 at depth 3), finds three depth-7 networks on 10 channels from two-layer prefixes by SAT that pass both verifiers, and proves depth 6 unsatisfiable for the same prefixes (depth 7 is optimal for 10 inputs [8]).

4.2 The search pipeline

The pipeline in `src/snt/` (C++20, no external dependencies) reimplements Wang’s method with additional controls. Its stages are sub-commands of one binary.

gen Generate-and-prune of reflection-symmetric prefixes layer by layer, with subsumption under channel permutations and under reflect-and-complement (the lemma of Bundala and Závodný [5]), keeping the k best prefixes by output-set size per layer when a `-keep` limit is given.

stack

Place a symmetric prefix on an outer block of channels and another on the inner block (*nested*), or a block and its reflection side by side (*mirrored*); the output set of the stack is the product of the block output sets.

extend

Greedy addition of one comparator at a time (a comparator together with its mirror image) to a new layer, keeping the 64 best partial layers by output-set size—Wang’s greedy sixth layer.

cnf The SAT encoding of Codish, Cruz-Filipe, Ehlers, Müller and Schneider-Kamp [4] as ported by Wang: comparator variables $g_{k,i,j}$ with mirror identification, used-channel variables, “one up / one down” propagation variables per input and channel, a per-input window encoding after a channel permutation that minimises the sum of windows (leading zeros and trailing ones of each prefix output become constants), the necessary last-layer constraints (only adjacent comparators in the last layer; span at most 3 in the second-to-last layer, with the implied structure), and the optional normal-form constraints ψ_1 and ψ_3 (no two adjacent idle channels in the last layer, and the corresponding forms for the second-to-last layer). Flags added for this work: `-no_nf` drops the normal-form constraints (the “necessary constraints only” model); `-open_last` leaves the last prefix layer partially specified for

the solver to complete; `-max_comps  $K$` adds a cardinality constraint on the number of suffix comparators (Sinz’s sequential counter [12]; a mirrored pair counts two); `-subnet  $w$` produces the Bundala–Závodný subnetwork relaxation that keeps only prefix outputs of window at most w .

decode

Read a model, undo the channel permutation by Knuth’s untangling (when a comparator would be reversed, swap the labels of its two channels for all later comparators; valid for a suffix because the prefix output set contains all sorted vectors), strip redundant comparators and re-layer.

prune

For each comparator (or each pair with `-pairs`), delete it and decide exactly by the output-set method whether the remaining network still sorts, caching the reachable set after each layer.

A driver script (`src/run_search.sh`) runs a whole experiment: CNF generation for the k best prefixes of a library, parallel solving with a time budget, decoding, and both verifiers on every decoded network. It writes `runs/<name>/log.txt` with instance sizes and per-instance outcomes and times. Every run reported below has such a log; the CNF and solution files themselves are not kept in the repository.

As a sanity check of the cardinality option: from the best two-layer prefix on 10 channels (9 comparators), a suffix of at most 22 comparators is satisfiable (size 31, matching the list entry (31, 7)) and a suffix of at most 20 is unsatisfiable.

4.3 Environment

All computation ran on a 4-core x86-64 virtual machine with 15 GB of RAM (a shared cgroup limit of 14.3 GB), Ubuntu 24.04, gcc 13.3, Python 3.11 with numpy, MiniSat 2.2.1 [11] from the distribution, and Kissat 4.0.4 and CaDiCaL 3.0.1 built from source. The machine was frequently shared between jobs, so all wall-clock times are upper bounds on what an idle machine would need; they are reported as logged.

5 Calibration on 28 channels

5.1 Reproduction of Wang’s pipeline (verified)

Wang’s repository was built from source with a plain Makefile against distribution libraries (its Bazel build could not fetch dependencies through the environment’s egress proxy), with one local patch: his output-set routine uses a 2^n -bit mask and needs about 15 GB at $n = 28$, so it was replaced by a sparse computation. His README pipeline was then run unchanged (Table 1).

Table 1: Wall-clock times of Wang’s pipeline, rebuilt and rerun on the machine of Section 4.3.

step	wall time
12-channel symmetric prefixes to depth 5 (keep 4 at the last layer)	13 min 47 s
16-channel symmetric prefixes to depth 5 (keep 1 per layer; two 83-output prefixes)	1 min 25 s
stacking (16 outer, 12 inner) and greedy sixth layer (keep 64)	11 s
window-minimising channel permutation and CNF generation for 8 prefixes	13 s
MiniSat on the 8 instances, 4 in parallel	20 min 47 s
decoding, stripping, re-layering	53 s

Solver times for the eight instances were 26.7, 39.0, 44.2, 45.8, 197.1, 549.8, 906.9 and 1202.5 s, all satisfiable. The eight decoded networks (28 channels, 13 layers, sizes 165, 167, 167, 168, 170, 170, 173, 173) are in `networks/n28d13_wang_pipeline_repro_1..8.txt` and were accepted by both verifiers on all 2^{28} inputs (`networks/VERIFICATION.txt`).

5.2 Validation of our encoder and choice of solver

Our own encoder was validated on Wang’s prefixes. His four best six-layer prefixes (928 outputs each), encoded by our `cnf` stage with a symmetric suffix of seven layers and solved by MiniSat, are satisfiable in 102, 286, 348 and 433 s, and the decoded networks (sizes 163, 166, 170, 173) pass both verifiers (`networks/n28d13_own_encoder_on_wang_prefix_0000..0003.txt`). Deleting channel 0 or channel 27 from each of the thirteen verified 28-channel networks (Wang’s own and our twelve) gives 27-channel 13-layer networks of size at least 154, so none of them improves the list’s 153; the best is kept as `networks/n27d13_by_channel_deletion.txt`.

On two of Wang’s instances (about 1000 prefix outputs, 7 layers, 50 MB of CNF each) MiniSat needed 27 and 39 s, Kissat 528 and 541 s, and CaDiCaL 813 and 1184 s on a loaded machine. MiniSat was used for every run below. Our 30- and 32-channel instances (150 000 to 180 000 variables, 5 to 6 million clauses) are two to four times larger than the largest instances solved in [4].

5.3 A greedy seventh layer destroys completability

Observation 1. Wang’s eight six-layer prefixes are all completable in seven further layers (Section 5.1). Extending each by one greedy layer chosen exactly as his sixth layer (comparator pairs added while the output set shrinks, keep 64) reduces the output sets from 928 to between 530 and 541. Asked for a reflection-symmetric completion of these seven-layer prefixes in six layers, MiniSat proves all eight instances unsatisfiable, in 4, 5, 5, 5, 5, 4, 8 and 9 s (`runs/calib_wang_L7_minisat`).

Small output sets are therefore not a proxy for completability past layer 6, and every fast negative in this paper that involved a greedy seventh layer (runs B, D, E, K30 and K32 below) is uninformative about the prefixes preceding that layer. The fair test of a six-layer prefix is Wang’s configuration itself—six fixed layers plus seven SAT layers—and that is the configuration used for every NEGATIVE label on a six-layer prefix.

6 A 27-input network with 152 comparators and 13 layers

6.1 Derivation

An exact single-comparator deletion test (`prune`, Section 4.2) was run on the published depth-13 networks for 22 to 28 inputs. Exactly one comparator on these seven networks is removable: comparator (23, 26) in layer 7 of the network listed as “27 inputs, 153 CEs, 13 layers”. Removing it leaves a 27-channel network with 152 comparators and 13 non-empty layers. The comparator is not redundant in the classical sense: it swaps its inputs on some outputs of the first six layers, so the network after layer 7 has a different output set with and without it, and only the exact test discovers that the last six layers sort both. After the deletion no further single comparator, and no pair of comparators (all 11 476 pairs tested), can be removed.

6.2 The network

Theorem 1. *The network of Figure 1 sorts 27 inputs. It has 152 comparators and 13 layers, with layer sizes 13, 13, 13, 13, 12, 13, 10, 10, 12, 10, 12, 10, 11. The number of distinct reachable binary vectors after each layer is 3 188 646; 272 160; 49 200; 11 424; 2 822; 884; 659; 472; 236; 118; 62; 39; 28, the last being the $n + 1 = 28$ sorted vectors.*

```

[(0,19),(1,26),(2,25),(3,24),(4,23),(5,22),(6,21),(7,20),(9,10),(11,17),(12,15),(13,14),(16,18)]
[(0,1),(2,3),(4,5),(6,7),(8,9),(11,16),(12,14),(13,15),(17,18),(19,26),(20,21),(22,23),(24,25)]
[(0,2),(1,3),(4,6),(5,7),(8,18),(9,14),(10,12),(13,17),(15,16),(19,24),(20,22),(21,23),(25,26)]
[(0,4),(1,5),(2,20),(3,21),(6,19),(7,25),(8,13),(9,17),(10,11),(12,15),(14,18),(22,24),(23,26)]
[(1,2),(3,19),(4,6),(5,22),(7,20),(8,10),(9,12),(11,13),(14,16),(15,17),(21,23),(24,25)]
[(0,8),(1,4),(2,6),(3,10),(5,7),(9,11),(12,13),(14,15),(16,17),(18,19),(20,22),(21,24),(23,25)]
[(1,9),(2,13),(4,8),(5,12),(6,10),(7,20),(14,24),(15,22),(17,25),(18,21)]
[(3,4),(6,14),(7,11),(8,15),(9,18),(10,17),(12,23),(13,21),(16,20),(19,26)]
[(1,3),(2,4),(5,6),(7,8),(10,13),(11,15),(12,16),(14,18),(19,24),(20,23),(21,22),(25,26)]
[(2,7),(4,8),(6,9),(10,11),(12,14),(13,15),(16,18),(17,21),(19,20),(23,24)]
[(2,3),(4,7),(5,6),(8,10),(9,12),(11,16),(13,14),(15,17),(18,19),(20,23),(21,22),(24,25)]
[(4,5),(6,7),(8,9),(10,12),(11,13),(14,16),(15,18),(17,19),(20,21),(22,23)]
[(3,4),(5,6),(7,8),(9,10),(11,12),(13,14),(15,16),(17,18),(19,20),(21,22),(23,24)]

```

Figure 1: The 27-input sorting network with 152 comparators and 13 layers, one layer per line, channels numbered from 0 (the format of Dobbelaere’s list). It is the list’s 27/153/13 network with comparator (23, 26) removed from layer 7. Files: `networks/n27d13_size152.txt` (with a provenance header) and `networks/n27d13_size152.json`.

6.3 Verification (verified)

The network was first verified on 23 September 2026 at 02:10 UTC (Verifier A 0.24s; Verifier B in both modes 2 min 39s). It was re-verified from a clean rebuild of Verifier A at 14:34 to 14:46 UTC the same day; the transcript below is verbatim from `runs/final_rerun/*.log`.

```

$ gcc -O3 -march=native -fopenmp -o src/verify_c src/verify_c.c
$ time src/verify_c networks/n27d13_size152.txt
file=networks/n27d13_size152.txt n=27 size=152 depth=13 inputs=134217728
RESULT: SORTS all 2^27 binary inputs (n=27, size=152, depth=13)
real 0m0.097s   user 0m0.350s   sys 0m0.004s   exit=0

$ time python3 src/verify_py.py --mode all networks/n27d13_size152.txt
file=networks/n27d13_size152.txt n=27 size=152 depth=13 modes=exhaustive,outputset
exhaustive: SORTS
outputset: SORTS (|output| per layer: [3188646, 272160, 49200, 11424,
2822, 884, 659, 472, 236, 118, 62, 39, 28])
RESULT: SORTS all 2^27 binary inputs (n=27, size=152, depth=13)
real 6m44.338s   user 3m7.000s   sys 3m32.963s   exit=0

$ sha256sum networks/n27d13_size152.txt networks/n27d13_size152.json
6344af2c06599c6215326b8cb1fd2dc67190a78fc05e00930404a39728a3963c  n27d13_size152.txt
39bb3b5ba3bb5ae46faa6d22bb500410667bd7db3b6ff2fdd87b00ed65505807  n27d13_size152.json

```

Both verifiers parse the file independently, agree on size 152 and depth 13, and accept it on all 134217728 binary inputs; Verifier B’s output-set mode confirms the result by an algorithm unrelated to input simulation. The JSON and text files contain identical layers. The test suites were rerun in the same session: `tests/test_verifiers.py` passed 4 of 4 tests in 37.8s and `tests/test_pipeline.py` passed 2 of 2 in 4 min 2.6s (Appendix B). This establishes Theorem 1.

6.4 No other network on the list has a removable comparator (negative)

Proposition 1. *Of the 54 networks on Dobbelaere’s list (version of 7 November 2025), exactly one has a removable comparator: the 27/153/13 network, from which exactly the comparator of Theorem 1 can be removed. After that removal nothing further is removable. No pair of comparators can be removed from the resulting 27/152/13 network (11476 pairs) or from Wang’s 28/159/13 network (12561 pairs).*

The single-deletion sweep is `runs/prune_all.log` (the 32-input network takes about four minutes); the pair tests are `runs/pairs_n27_152.log` and `runs/pairs_n28_159.log`. Each

deletion is decided by the exact output-set method, so the proposition is a computed fact, not a heuristic finding.

6.5 Size at depth 13 for 27 and 28 inputs by SAT with a cardinality bound

Let P_0 be Wang’s first six-layer 28-channel prefix (83 comparators, 928 outputs); his 159-comparator network is P_0 plus a 76-comparator suffix. Table 2 lists the SAT runs with a cardinality bound on the suffix.

Table 2: Size-bounded completions. All runs: MiniSat, 7 SAT layers, encoding of Section 4.2.

run	model	result	label
S28	P_0 + 7 symmetric layers, at most 75 suffix comparators, with normal forms	UNSAT, 989 s	NEGATIVE under normal forms only (ψ_1 can force redundant comparators into the count)
S28b	same, necessary constraints only (<code>-no_nf -max_comps 75</code>)	UNSAT, 1207 s	NEGATIVE (Proposition 2)
S28c	same with a non-symmetric suffix	timeout, 14 400 s	INCONCLUSIVE
S27	first 6 layers of the 27/152 network (77 comparators, 884 outputs) + 7 non-symmetric layers, at most 74 suffix comparators (total ≤ 151), necessary constraints only	timeout, 14 400 s	INCONCLUSIVE

Proposition 2. *No reflection-symmetric comparator network of 7 layers with at most 75 comparators sorts every output of P_0 . Consequently 159 is the minimum size of a reflection-symmetric 13-layer sorting network on 28 channels that begins with P_0 .*

The statement is sound because the non-redundant form of any 13-layer completion satisfies the necessary last-two-layer constraints, is no larger, and remains reflection-symmetric (Section 7.4). Whether a 27-input 13-layer network with 151 comparators, or a 28-input one with 158, exists remains open.

7 Negative results for 30 channels

7.1 Prefix families

All prefixes are reflection-symmetric on 30 channels. VV16 denotes the first five layers of Van Voorhis’s 16-input network as used by Wang (a 4-cube followed by a fifth layer, 83 outputs; two variants exist with the same output-set size). Block prefixes on 14 channels were generated with the `gen` stage under three regimes: keep 1 per layer (69 outputs, 8 s); keep 4, 4, 4, 8 (eight prefixes with 66, 69, 71 and five times 72 outputs, 9 s); and a keep-limited library (all 69 prefixes at depth 2; 2000 of 16 119 locally pruned candidates kept at depth 3; 4389 of 49 487 at depth 4 from 5.76 million generated candidates, about 9.6 hours of wall clock including pauses; 32 prefixes at depth 5, best 66 outputs, 6450 s for the last step). The library’s best value equals the greedy keep-4 value, so the more exhaustive generation did not improve the stacks. Table 3 summarises the families.

For comparison, Wang’s 28-channel prefixes have $2905 = 83 \times 35$ outputs after five layers and 928 after the greedy sixth layer. According to the literature review recorded in `NOTES.md`, every published SAT completion that left seven layers to the solver started from at most about 1200 prefix outputs (Codish et al. about 800 and 840 on 17 channels, Ehlers 1154 on 24 channels, Wang 928 on 28 channels). The best 30-channel six-layer prefixes here have 1693 to 1853.

Table 3: 30-channel prefix families. $|\text{out}|$ after five layers, after the greedy sixth layer (keep 64) and, where computed, after a greedy seventh layer.

family	construction	5 layers	+ greedy 6th	+ greedy 7th
S	VV16+VV16 nested on 32 channels, channels 0 and 31 deleted	6723	1699 (65 prefixes)	928 (74)
D30	first layers of the list’s 30/172/14 network	7413	2564 (its own 6th)	1033 (its own 7th)
C30	5-cube with channels 0 and 31 deleted	7579	—	—
N1	VV16 (83) + keep-1 14-channel prefix (69), both nestings	5727	1983 (82)	—
N4	VV16 + keep-4 14-channel prefixes, both nestings, 32 stacks pooled	5478–5976	1853 (66 prefixes; 1853–1973)	—
NL	VV16 + library 14-channel prefixes, both variants and nestings, 128 stacks	5478–6059	1693 (best 8: 1693, 1693, 1697, 1697, 1729, 1729, 1733, 1733)	—
structured	N4 stacks + Ehlers-style rank-aligned cross-block 6th layer	5478	3001–3689	—

7.2 SAT runs

Table 4 lists every run. The suffix is required to be reflection-symmetric unless stated; “SAT layers” is the number of layers left to the solver. Per-instance CNF sizes and times are in Appendix A.

Run A’s timeouts were CPU contention: the same two instances are UNSAT in 549 and 909 s when rerun with two jobs (A2). The Bundala–Závodný subnetwork relaxation was tried as a cheap pre-screen on run F’s first prefix: with window at most 8 it is satisfiable in 2.3 s, with window at most 12 in 28.8 s, and with window at most 16 it did not finish in ten minutes; it therefore does not filter these instances (`runs/subnet_screen`).

7.3 Statement

Theorem 2. *For each of the 22 six-layer prefixes of runs A2, F, F5, F’ and F’’ (2 from family S, 12 from N4, 8 from NL), there is no reflection-symmetric comparator network of 7 layers that sorts every output of the prefix. Hence none of these prefixes extends to a reflection-symmetric 13-layer sorting network on 30 channels.*

Each proof of unsatisfiability took between 500 and 1800 s.

7.4 Soundness of the negatives

Each NEGATIVE label rests on two points.

1. The CNF is a correct model of “a symmetric suffix of the given depth sorts every prefix output”. This was validated positively: the same encoder produces satisfiable instances and verified networks on Wang’s 28-channel prefixes (Section 5.2) and finds the depth optimum on 10 channels (Section 4.1).
2. The restrictions on the last two layers (last layer adjacent comparators only; second-to-last layer span at most 3 with its implications; Lemmas 4 and 6 of [4]) are necessary for the non-redundant form of any sorting network of the given depth, and removing redundant comparators preserves reflection symmetry, because a comparator and its mirror image are

Table 4: SAT runs on 30 channels (MiniSat 2.2). Runs marked “uninformative” involve a greedy seventh layer (Observation 1).

run	prefixes (out)	SAT layers	budget	outcome	label
B	S, 7 layers (928–940), best 8	6	1800 s	8/8 UNSAT, 8–12 s	NEGATIVE, uninformative
D	S, 6 layers + 3 greedy layer-7 pairs frozen (1186–1208), best 8	6	1800 s	8/8 UNSAT, 15–20 s	NEGATIVE, uninformative
E	as D, layer 7 open for the solver on the other 24 channels	6+	1800 s	8/8 UNSAT, 15–24 s	NEGATIVE, uninformative
A	S, 6 layers (1699, 1735, 1915, 1915)	7	1800 s	4/4 timeout; prefixes 5–8 not run	INCONCLUSIVE
A2	S, 6 layers, prefixes 1 and 2 (1699, 1735)	7	7200 s	2/2 UNSAT, 549 and 909 s	NEGATIVE
F	N4, 6 layers, best 4 (1853, 1853, 1857, 1857)	7	7200 s	4/4 UNSAT, 501, 512, 517, 520 s	NEGATIVE
F-control	N4 prefix 1 (1853), normal forms removed (<code>-no_nf</code>)	7	14 400 s	UNSAT, 654 s	NEGATIVE
F5	N4 prefixes 5–12 (1873–1893)	7	1200 s	8/8 UNSAT, 841–1192 s	NEGATIVE
F'	NL, best 8 (1693–1733)	7	1200 s	prefixes 1–4 UNSAT, 798–823 s; 5–8 timeout	NEGATIVE (4), INCONCLUSIVE (4)
F''	NL prefixes 5–8 (1729, 1729, 1733, 1733)	7	7200 s	4/4 UNSAT, 1650, 1634, 1057, 1078 s	NEGATIVE
H	N4 prefix 1 (1853), suffix not required to be symmetric	7	14 400 s	timeout	INCONCLUSIVE
G	N4, 5 layers (5478, 5478), no greedy layer	8	14 400 s	killed by a container restart after about 3 h	INCONCLUSIVE
K30	D30, 7 layers (1033)	6	3600 s	UNSAT, 3 s	NEGATIVE, uninformative

redundant together. They therefore exclude no symmetric completion. The normal-form constraints ψ_1 and ψ_3 are proven in [4] for unrestricted networks; rather than argue their soundness under the symmetry restriction, the control run re-proved run F’s first instance with them removed (654 s instead of 520 s). The NEGATIVE labels for the other 21 prefixes still include the normal-form constraints; a reviewer who wants to remove that dependence can rerun any of them with `-no_nf` at about the same cost.

What the negatives do not say: nothing about non-symmetric completions (run H timed out), nothing about other sixth layers on the same five-layer stacks (run G was killed; the structured sixth layers were not solved), and nothing about prefixes outside these families.

8 Negative results for 32 channels

Seed prefixes (five layers, reflection-symmetric): VV16+VV16 nested or mirrored ($6889 = 83^2$ outputs; the two arrangements have identical output sets), the first five layers of the list’s 32/185/14 network (6887), the 5-cube (7581), and two blocks of the list’s 61-comparator 9-layer 16-input network side by side (12 100). The greedy sixth layer reduces VV16+VV16 to 1787 (64 prefixes) and a greedy seventh to 992 (83 prefixes); it reduces the 5-cube to 2416 (51 prefixes; 69 minutes under load). Ehlers-style structured sixth layers on VV16+VV16 give 2540 (translation or coordinate pairing); reflection-pair layers are much worse (4699 to 6561 across the 30- and 32-channel stacks). Table 5 lists the runs.

Theorem 3. *None of the four best greedy six-layer prefixes of the Van Voorhis 16+16 family, and none of the two best greedy six-layer prefixes of the 5-cube family, extends to a reflection-symmetric*

Table 5: SAT runs on 32 channels (MiniSat 2.2, symmetric suffix).

run	prefixes ($ \text{out} $)	SAT layers	budget	outcome	label
K32	first 7 layers of 32/185/14 (1231)	6	3600 s	UNSAT, 4 s	NEGATIVE, uninformative
F32	VV16+VV16 nested, 6 layers, prefixes 1 and 2 (1787)	7	7200 s	2/2 UNSAT, 812 and 798 s	NEGATIVE
F32b	same family, prefixes 3 and 4 (1823)	7	7200 s	2/2 UNSAT, 1790 and 1793 s	NEGATIVE
C32	5-cube + greedy 6th layer, best 2 (2416, 2516)	7	3600 s	2/2 UNSAT, 818 and 779 s	NEGATIVE

13-layer sorting network on 32 channels.

Not attempted (INCONCLUSIVE): eight free layers on the five-layer VV16+VV16 stack; non-symmetric suffixes; cube-and-conquer. Since no 32-channel network was found, nothing follows for 31, 30 or 29 channels by deletion, and a direct search on 29 channels, which needs a relaxed symmetry (Section 2), was not attempted.

9 Discussion

Why the transfer fails where Wang’s succeeded. Two quantitative differences stand out. First, the starting point: after five layers Wang’s stacks have 2905 outputs, ours 5478 to 7581, because the 14-channel and second 16-channel blocks are far from the 35-output quality of his exhaustively optimised 12-channel block. The best five-layer 14-channel symmetric prefix we could produce has 66 outputs; a 14-channel block would need fewer than 81 outputs merely to beat the seed family’s 6723 and far fewer to match Wang’s ratio. Second, the hand-over to SAT: after the greedy sixth layer we hand the solver 1693 to 1853 outputs, where every successful completion in the literature handed it at most about 1200. Whether this is a barrier or an artefact of the greedy sixth layer is the central open question. Ehlers’ 24-channel result is instructive: according to our reading of his account (recorded in `NOTES.md`), his completable prefix had a SAT-chosen rank-aligned sixth layer (1154 to 479 outputs) that a comparator-at-a-time greedy does not find, and he observes that greedy prefix construction typically fails because of a few bad decisions, suggesting limited-discrepancy search around the greedy choice.

Block completable does not discriminate. We tested whether each five-layer block prefix completes, under symmetry, to a depth-optimal sorter of the block alone (16 channels in depth 9, 14 in depth 9, 12 in depth 8). The two 83-output VV16 variants split (one is symmetric-completable, the other only non-symmetrically), the 14-channel prefixes split, and among Wang’s eight successful prefixes the inner 16-blocks split four to four. A depth-13 network on 32 channels cannot first sort two 16-blocks (9 layers each, in parallel) and then merge them (at least 5 layers, since the middle output depends on all 32 inputs): sorting and merging must interleave, and in Wang’s network the cross-block comparators begin in layer 6.

What a follow-up should try , ranked by expected value per CPU hour:

1. Larger budgets for the fair instances. Per our reading of the literature, Codish et al. spent up to about 97 000 s on single instances and Bundala and Závodný up to 24 hours; our budgets of 1800 to 7200 s are small for instances two to four times larger. Run G (five-layer stack, eight free layers, 602 000 variables and 21.8 million clauses) and run H (non-symmetric suffix) are the natural candidates; on four cores that is days.

2. A SAT-scored beam or limited-discrepancy search for layer 6 instead of greedy minimum output set: keep the 200 to 500 best sixth-layer candidates, sweep them with a 900 s budget each, and invest hours only in survivors.
3. Better block prefixes: a true optimum for 14 channels (our libraries were keep-limited), blocks from the 61-comparator 16-input network (110 outputs but a different structure), a beam-searched sixth layer on the 5-cube for 32 channels, and 15+15 mirrored stacks once a fast non-symmetric 15-channel generator is available.
4. Non-symmetric suffixes throughout (the instance doubles; run H is the template).
5. Encoding: add the co-saturation constraints of [4] (their Lemma 8) and the solver settings used there (probing preprocessing, clause decay 0.9999).
6. Cube-and-conquer on the layer-7 variables of the best instance, to use many cores.

10 Limitations

- All negatives are for reflection-symmetric suffixes and specific prefixes; they do not bound depth 13 for 30 or 32 inputs in general.
- The normal-form constraints were removed in one control instance only; the other 21 negatives on 30 channels and all negatives on 32 channels include them.
- Wall-clock times were measured on a shared four-core machine and are not comparable to each other or to the literature at better than a factor of two; the run A timeouts show the effect.
- The 14-channel prefix libraries were keep-limited, not exhaustive; a better 14-channel block may exist.
- The list entries used as baselines are those of 7 November 2025; a later update of the list would change the improvement claim, not the network.

11 Reproducibility and data availability

All material is in the directory `sortnet_d13/` of the repository. The commands below are relative to it.

```
gcc -O3 -march=native -fopenmp -o src/verify_c src/verify_c.c
src/verify_c networks/n27d13_size152.txt # 0.1 s
python3 src/verify_py.py --mode all networks/n27d13_size152.txt # ~7 min
python3 tests/test_verifiers.py # ~40 s
make -C src/snt && python3 tests/test_pipeline.py # kissat in PATH; ~4 min
src/snt/snt prune --net data/dobb_nets/N27L153D13.txt --multiline
src/snt/snt prune --net networks/n27d13_size152.txt --multiline --pairs
src/run_search.sh <name> <library> 13 0 64 minisat <budget> <count> <jobs> \
    [--no_nf] [--max_comps K]
```

The prefix libraries of every run are under `runs/<run>/prefix_in.txt`, the per-instance sizes and times in `runs/<run>/log.txt`, the list snapshot in `data/`, Wang’s eight six-layer prefixes in `data/prefixes/wang_n28d6_64.txt`, the chronological log with timestamps in `NOTES.md`, and the full run tables in `results.md`. A reviewer with limited time should run the two verifier commands, then `tests/test_verifiers.py`, then one 500 s instance of run F with and without `-no_nf`.

12 Conclusion and confidence

The claim that the 152-comparator network sorts rests on two independent exhaustive checks of all 2^{27} binary inputs by programs written separately, one of which also confirms the result by

output-set propagation, and both of which were validated against every network of Dobbelaere’s list with $n \leq 20$ and against deliberately broken networks. The depth is the number of non-empty layers in the listing (13) and the size the number of comparators in it (152); neither can be reduced by stripping, single deletion or pair deletion. The improvement claim rests on the list entry (153, 13) for 27 inputs as of 7 November 2025.

The reproductions of Wang’s bound are equally solid and add nothing new. The NEGATIVE results are exact statements under a stated model whose two premises—a correct encoding and sound last-layer restrictions—were each tested positively; the one residual dependence, on the normal-form constraints under symmetry, was removed for one instance and can be removed for the others at known cost. Whether a 13-layer sorting network exists on 29, 30, 31 or 32 channels remains open. The evidence here says only that the most direct transfer of the one method known to reach depth 13 beyond 26 channels does not work with the block prefixes and greedy sixth layers we could produce, and that the next attempt should change the sixth layer, the blocks, or the budget rather than repeat these runs.

A skeptical reader should rerun the two verifier commands of Section 11 first, then the control instance of run F.

Credit. Suggested credit line for the list entry: “[Author name], with an AI research session (Claude)”. A draft submission note for the list maintainer is in `submission_email.txt`; it has not been sent.

References

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A Instance sizes and per-instance outcomes

All instances: MiniSat 2.2; variable and clause counts from the CNF header as logged; times are wall-clock seconds as logged. Run A (eight instances, 1699 to 1977 outputs, 147 913 to 178 249 variables, 1800 s budget, four jobs) produced four timeouts and was superseded by A2. The H instance’s variable count (about 325 000, twice the symmetric instance) is from the run description in NOTES.md; its CNF header line was not preserved in the log.

run	inst.	prefix outputs	variables	clauses	outcome (s)
A2	0	1699	147 913	4 998 046	UNSAT 549
A2	1	1735	150 569	5 085 804	UNSAT 909
F	0	1853	159 353	5 246 802	UNSAT 520
F	1	1853	159 577	5 267 242	UNSAT 501
F	2	1857	160 153	5 302 918	UNSAT 512
F	3	1857	160 409	5 324 598	UNSAT 517
F-control	0	1853	159 353	5 246 717	UNSAT 654
F5	0–7	1873 ($\times 4$), 1889 ($\times 2$), 1893 ($\times 2$)	160 777–162 473	5 203 072–5 390 088	UNSAT 841, 1112, 1122, 1192, 1032, 985, 1000, 880
F′	0–3	1693, 1693, 1697, 1697	146 361–147 449	4 867 514–4 963 014	UNSAT 798, 815, 807, 823
F′	4–7	1729, 1729, 1733, 1733	148 985–150 105	4 956 888–5 050 772	timeout 1200
F′′	0–3	1729, 1729, 1733, 1733	149 113–150 105	4 979 656–5 050 772	UNSAT 1650, 1634, 1057, 1078
H	0	1853	about 325 000	not logged	timeout 14 400
G	0, 1	5478	601 982	21 832 822 / 21 838 246	killed after about 3 h
B	0–7	928 ($\times 4$), 940 ($\times 4$)	64 978–66 602	1 955 238–2 087 270	UNSAT 11, 11, 10, 12, 9, 10, 9, 8
D	0–7	1186–1208	87 924–91 844	2 873 078–3 126 050	UNSAT 16, 18, 20, 18, 20, 18, 16, 15
E	0–7	1186–1208	100 585–105 065	3 347 772–3 642 742	UNSAT 18, 17, 24, 20, 21, 19, 18, 15
K30	0	1033	58 398	1 366 654	UNSAT 3
K32	0	1231	81 284	2 199 176	UNSAT 4
F32	0, 1	1787	159 874 / 160 098	5 583 368 / 5 604 228	UNSAT 812, 798
F32b	0, 1	1823	162 562 / 162 786	5 662 622 / 5 682 390	UNSAT 1790, 1793
C32	0, 1	2416 / 2516	183 314 / 186 946	4 994 662 / 5 033 048	UNSAT 818, 779
S28	0	928	271 628	2 654 064	UNSAT 989
S28b	0	928	271 628	2 653 985	UNSAT 1207
S28c	0	928	275 842	2 682 405	timeout 14 400
S27	0	884	253 910	2 393 272	timeout 14 400
calib, own encoder on Wang prefixes	0–3	928	73 178–73 306	2 221 386–2 257 966	SAT 102, 433, 286, 348
calib + greedy 7th layer	0–7	530 ($\times 5$), 540, 541, 541	34 918–36 528	961 960–1 082 112	UNSAT 4, 5, 5, 5, 5, 4, 8, 9

B Test-suite transcript (rerun of 23 September 2026)

```
$ time python3 tests/test_verifiers.py
PASS test_dobbelaere_accept_and_broken_reject
PASS test_parse_errors
```

```

PASS test_trivial_networks
PASS test_wang28
4/4 passed
real 0m37.817s  exit=0

```

```

$ time python3 tests/test_pipeline.py
PASS test_gen_counts_n12
PASS test_sat_chain_n10
2/2 passed
real 4m2.555s  exit=0

```

C Verified networks in the repository

Every file below was accepted by both verifiers on all 2^n binary inputs; the outputs are recorded in `networks/VERIFICATION.txt`.

file (in <code>networks/</code>)	n	depth	sizes	origin
<code>n27d13_size152</code>	27	13	152	the list's 27/153/13 with layer-7 comparator (23, 26) deleted (Theorem 1)
<code>n27d13_by_channel_deletion</code>	27	13	154	best channel deletion from the 28-channel networks below
<code>n28d13_wang_pipeline_repro_1..8</code>	28	13	170, 167, 167, 165, 173, 173, 170, 168	Wang's pipeline, rebuilt and rerun
<code>n28d13_own_encoder_on_wang_prefix_0000..0003</code>	28	13	170, 166, 173, 163	our encoder on Wang's prefixes